# ELI — Dag

ELI v2.5.32

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## ELI DAG Engine — project-wide dependency scheduling

A single, generic directed-acyclic-graph engine (`eli/core/dag.py`) is now the shared scheduling primitive across ELI: the **agent bus** runs its agents on it, the **coding engine** runs its subtasks on it, and **MULTI_COMMAND** orders chained commands on it. One DAG algorithm, three consumers — no per-subsystem reinvention.

### The engine (`eli/core/dag.py`)

Pure, deterministic, no I/O — fully unit-tested.

- `DAG` / `DAGNode` — nodes with `depends_on` edges (B depends_on A ⇒ A→B).
- `topological_order()` — Kahn's algorithm, deterministic (sorted ties); raises `DAGCycleError` on a cycle.
- `topological_layers()` — groups nodes into **parallel layers**: layer *k* holds every node whose dependencies are all in earlier layers. This is what lets independent work run concurrently while dependent work waits.
- `ancestors()` / `descendants()` — transitive closure.
- `critical_path_length()` — longest dependency chain = number of sequential layers (a parallelism metric).
- `build_dag({node: [deps]})` — convenience builder that **drops deps outside the given node set**, so a global dependency map can be applied to any subset safely.

### Consumer 1 — the agent bus runs on the DAG (`eli/cognition/agent_bus.py`)

Previously the bus was a single-round star fan-out (agents couldn't use each other's output). Now:

- `_AGENT_DEPENDENCIES` declares edges among agents. Shipped edges: `knowledge_graph ← memory`, and `critic ← {memory, system, knowledge_graph, file_code}` — `CriticAgent` is the bus's second-tier verifier: it reasons over what its upstream retrievers found (via `intent["_upstream"]`) rather than the raw query, measuring pairwise term-overlap agreement between them and downgrading

confidence on contradiction. It self-gates to nothing when fewer than 2 upstream sources are available, so it never fires on trivial turns.

- `_agent_execution_layers(active_names)` builds a DAG from that map ∩ the selected agents and returns topological layers.
- `AgentBus._run_agents_layered` runs **layer by layer**: each layer's agents run in parallel (same per-agent hard timeout + failure handling as before), then every completed layer's results are passed to downstream agents via `intent["_upstream"]`. The `knowledge_graph` agent uses this to seed its lookup with what `memory` surfaced this turn.
- **Non-regressive:** when no edges apply to the selected set, the DAG collapses to a single layer = the legacy flat fan-out. Gated by `ELI_AGENT_DAG` (default on); any error falls back to flat dispatch (`_run_agents_flat`).

```
selected agents → _agent_execution_layers → [[memory, reflection, …], [knowledge_graph], …]
                                                 | run parallel       | receives _upstream={memory:…}
```

Add an edge by editing `_AGENT_DEPENDENCIES`; cycles are rejected at build time.

## Consumer 2 — the coding engine's subtask DAG (`eli/coding/plan_graph.py`)

For decomposable coding tasks:

- `decompose_dag(task, generate)` asks the planner for a **graph** of subtasks (`{nodes:[{id, task, depends_on}]}`), builds a `DAG`, and validates it (cycle/ dangling-edge safe). A cohesive task yields a single node.
- `solve_dag(...)` walks the topological order, solving each node via the normal single-shot tree search and **feeding each node the code of its dependencies**, then `compose()` concatenates the node solutions (imports hoisted/deduped) into one module. `CodeAgent.solve` verifies the composed module against tests synthesized for the *original* task.
- Single-node tasks return `None` so the agent uses the plain single-shot path (no DAG overhead). Gated by `ELI_CODING_DAG` (default on).

## Consumer 3 — MULTI_COMMAND's dependency graph (`eli/execution/command_dependency_g` 2026-09-29)

For a chained command request ("close steam and set an alarm" / "open the file and then edit it"):

- `infer_dependencies(commands)` mirrors `plan_graph.decompose_dag`'s shape: one optional LLM call asking only for dependency EDGES among the fixed, already-split command list (not a re-decomposition), validated through `build_dag(...).topological_layers()`. Falls back to fully independent (empty dict — today's behavior) on no model, bad JSON, an out-of-range index, or a cycle.
- The executor runs `MULTI_COMMAND` in topological layers. A step whose dependency didn't succeed is skipped and reported as such (not silently run); independent branches still run regardless of an unrelated failure. Gated by `ELI_MULTI_COMMAND_DAG` (default on).
- Deliberately narrower than consumers 1-2: it only asks for edges among an already-fixed list, never regenerates or reorders the commands themselves — the router's `command_splitter.py` owns splitting, this only owns ordering.

## Scope & honesty

- **DAG-driven now:** agent-bus execution (topological layers + upstream), coding subtask decomposition, and MULTI_COMMAND dependency-aware execution all run on `eli/core/dag`.
- **Still linear (not yet DAG):** the 12-stage orchestrator pipeline is a fixed chain; `execution_planner.ExecutionPlan.steps` is an ordered list. Converting those to DAGs is a possible next step but wasn't required for "agents on a DAG".
- **Composition is v1:** the coding DAG composes node code by ordered concatenation with import-deduping; integration quality scales with the model.

- Everything here is covered by `tests/test_dag.py` (engine, layered dispatch via fake agents, coding subtask DAG) plus `tests/test_multi_command_dependencies.py` (consumer 3) — all deterministic, no real model calls in the test suite.

## Control summary

| Knob | Default | Effect |
|---|---|---|
| `ELI_AGENT_DAG` | on | agent-bus topological layered dispatch (off ⇒ flat fan-out) |
| `ELI_CODING_DAG` | on | coding subtask-DAG decomposition (off ⇒ single-shot only) |
| `ELI_MULTI_COMMAND_DAG` | on | MULTI_COMMAND dependency inference (off ⇒ fully independent) |
| `_AGENT_DEPENDENCIES` | `{knowledge_graph: {memory}, critic: {memory, system, knowledge_graph, file_code}}` | declare agent edges |